Professional Program in

Data Science, Machine Learning, AI & GenAI

eict IIT Roorkee

E&ICT, IIT Roorkee

The IoT Academy
Connecting The Unconnected

Brochure

Duration: 10 Months | 240 Hours

DS | ML | DL | AI | Generative AI

Advanced Gen AI, Agentic AI & LLMs Are Covered



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About The IIT Roorkee

Indian Institute of Technology Roorkee (IIT Roorkee), one of the oldest technical institutions in Asia, was established in **1847** as the Thomason College of Civil Engineering and was later converted to an IIT in 2001

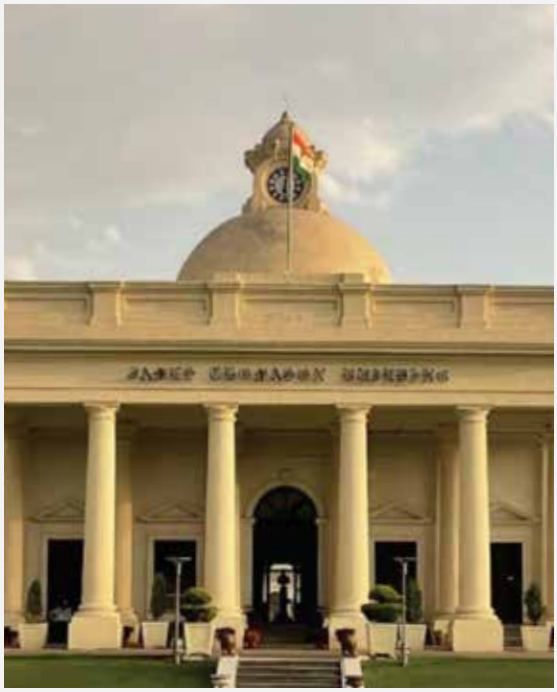


The institute offers a wide range of academic programmes including BTech, BArch, BS, MTech, MSc, MBA, MDes, and PhD across its 23 academic departments, centres, and schools covering engineering, sciences, architecture, management, humanities, and social sciences.

Over the years, **IIT Roorkee** has developed state-of-the-art infrastructure, a strong research ecosystem, and a rich legacy of excellence in technical education and innovation. The institute continues to be recognized nationally and internationally for its academic rigor, research output, and contributions to industry and society.

About The E&ICT Academy IIT Roorkee

Electronics & ICT Academy, IIT Roorkee aims to strengthen the skills of faculty members, working professionals, and students from Engineering, Science, Commerce, and Polytechnic institutions through specialized short-term training programs.
The Academy focuses on both fundamental and advanced areas of Information Technology, Electronics & Communication, Product Design, and Manufacturing.



In addition, E&ICT Academy, IIT Roorkee organizes customized industry-oriented training programs, faculty development programs (FDPs), and research promotion workshops to bridge the gap between academia and industry.
These initiatives are designed to enhance technical competency, encourage innovation, and promote collaborative research in emerging technologies.

About The IoT Academy

The IoT Academy, established in 2017 is a fast emerging company imparting quality programs for skills training, internship and guidance in cutting edge technologies like Data Science, Machine Learning, Artificial Intelligence, Internet of Things, Embedded Systems & many more and focused on helping people develop the skills they need to thrive in the rapidly growing digital economy.

The IoT Academy has collaborated with various premier institutes e.g. E&ICT Academy, IIT-Roorkee, IIT-Guwahati and IIT-Kanpur for Advance Certification courses to take provide outcome-centric solutions to help them achieve their professional goals.

400+
Hiring Partners

30,000+
Learners

8+ Years
Of excellence

81%
Salary Hike

91%
Happy Outcomes





About The Program

Most of the modern technological innovations you see today are powered by Data Science, Machine Learning, AI, and Generative AI. This 10-months Professional Program In Data Science, Machine Learning, AI & GenAI is thoughtfully designed by E&ICT Academy, IIT Roorkee, in collaboration with industry experts, to help you master the exact skills the market demands. The primary goal of this certification program is to prepare you for the most in-demand roles in Data Science, Machine Learning, AI, and Generative AI. You will gain expertise in Artificial Intelligence & Data Technologies, Advanced Analytics, Intelligent Systems, Machine Intelligence, Prompt Engineering and AI/ML-driven Solutions through multiple real-world projects in Machine Learning, Deep Learning, and Generative AI — transforming you into a highly skilled and industry-ready New Age DS, ML, AI & Gen AI Specialist.

Integrated With Gen AI, Agentic AI & LLMs

- Gen AI & Agentic AI Modules & Architecture
- Building LLM Applications
- Gen AI & Agentic AI Tools & Case Studies

Why Learners Choose Us Over the Rest?

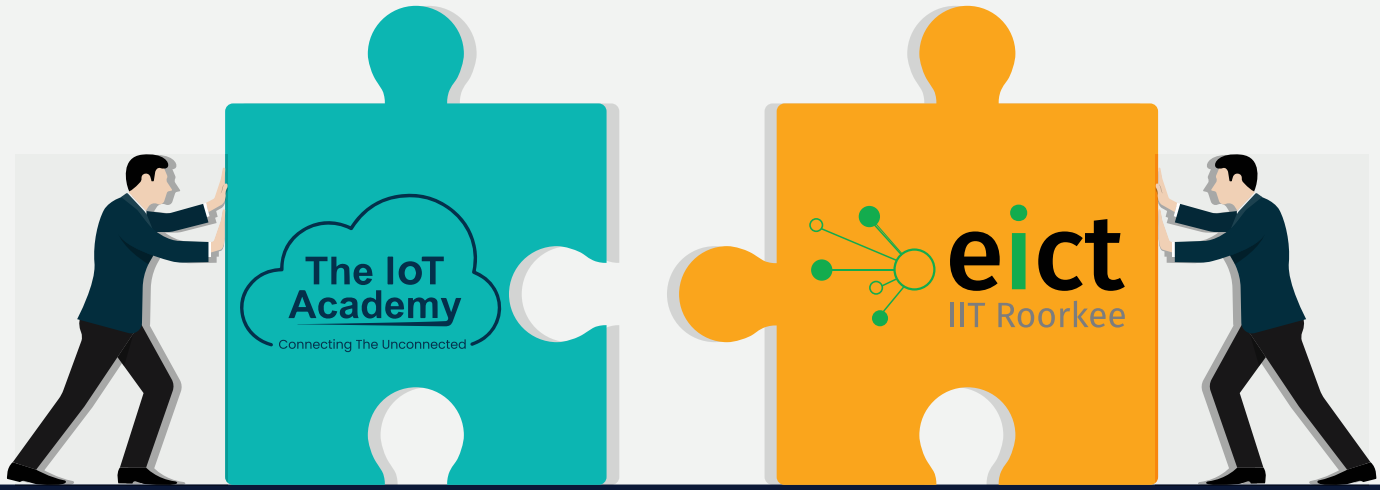
Program Perks	Live Learning For Real-Time Experience and Classroom Feel	Other Institution
Only Program With DS, ML, AI & GenAI	✓	✗
Online Flexible Learning	✓	✗
1:1 Live Doubt Session	✓	✗
Weekend Classes	✓	✗
Taught by IIT Faculty	✓	✗
Recorded Lectures Lifetime Access	✓	✗
Applied Industry Projects	✓	✓
Live Capstone Projects	✓	✓
Resume & Interview Preparation	✓	✓
3 Month Industrial-Internship	✓	✗
Industry Partner's Alumni Network Access	✓	✗

PARTNERSHIP



About The Partnership

Professional Program In Data Science, Machine Learning, AI & GenAI is offered jointly by The IoT Academy and E&ICT Academy, IIT Roorkee to prepare the future workforce in Machine Learning and AI.





Why Are Data Science, Machine Learning, AI and Gen AI Important?

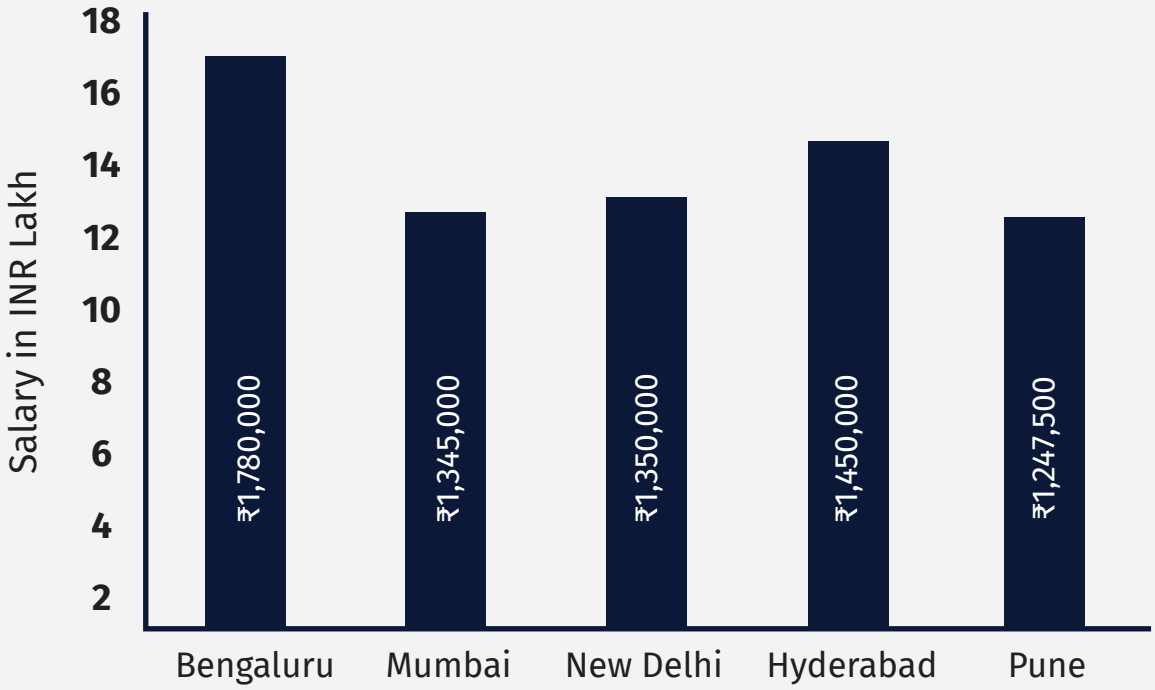
Data Science, Machine Learning, AI, and Generative AI are transforming the world around us, enabling machines to think, learn, and create in ways once possible only for humans. These technologies are driving innovation, reshaping industries, and redefining how we live and work. The global AI market is projected to reach $64.8B by 2028, growing at an annual rate of 43%. In terms of revenue, the global Data Science and ML market is expected to surpass US $5.8B by 2030, while companies are forecasted to invest up to $7.2 Trillion in AI and Gen AI by 2028. The total economic impact of AI and Generative AI is estimated to range between $6 and $15 Trillion per year by 2030, marking it as one of the most powerful growth drivers of the future.

- Virtual Personal Assistants
- Search Engine Result Refining
- Social Media Services
- Online Fraud Detection
- Email Spam and Malware Filtering
- Product Recommendations

Current Trends In DS, ML AI Job Market

As per Gartner, Data Science Machine Learning and AI will create 23 Lakhs job openings by 2026-27 The average annual salary of an DS, ML-AI Engineer in India is:

- **Freshers:** Between ₹ 6,99,807-8,91,326 (Salary per annum).
- **Experienced:** Between ₹ 35,00,000-50,00,000 (Salary per annum).



Source- glassdoor



Program Highlights

This 10-months program is uniquely designed by E&ICT Academy, IIT Roorkee, and industry leaders to help you master exactly what today's job market demands.

- Certification from E&ICT Academy, IIT Roorkee
- 100% Placement Assistance provided by The IoT Academy
- Hands-on learning with real-world Industry-Grade Capstone Projects
- Gain industry-relevant skills in Data Science, Machine Learning, AI & Generative AI
- Curriculum designed by leading academic and industry experts along with E&ICT Academy, IIT Roorkee faculty

Program Benefits



Practical Knowledge

Engaging, live instructor-led online classes conducted by experienced industry experts.



Personalised Attention

Dedicated technical and non-technical teams to assist and resolve all your queries.



Curriculum

Expertly designed by E&ICT Academy, IIT Roorkee professors & leading industry professionals.



Doubt Clearing Sessions

24x7 support from subject matter experts for all your technical and non-technical doubts.



Support and Guidance

Lifetime access to our Learning Management System and continuous program support.



Certificate/Credentials

Earn a prestigious Certificate of Completion from E&ICT Academy, IIT Roorkee.



Career Guidance & 100% Placement Support By Industry Partner

Full access to our Career Assistance and Placement Services.



Immersive Learning Experience

A next-gen learning ecosystem designed to provide an offline-like, interactive experience online.

Key Features

Learn from one of the best faculty in India through live online sessions. While Academic Professors will help you learn data science concepts, industry experts will impart practical knowledge of data science, machine learning, deep learning, and AI techniques through real-world projects.

Professional Program In
Data Science, Machine Learning, AI & GenAI

By E&ICT Academy, IIT Roorkee

- **10-Months Online Learning**
- **Project Work**
- **Live Ask Expert Anything Sessions & Code**
- **Code Reviews & Personalized Feedback**
- **Doubt Resolution on LMS**
- **Live Test, Home Assignments**
- **8-10 Hours Per Week Live Instructor**



Course Fee



eict IIT Roorkee

Professional Program In Data Science, Machine Learning, AI & GenAI

₹ 95,000 + 18% GST

Avail Scholarship Up To 30%

Financial Partner

Candidates can pay the course fee through Netbanking and Credit/Debit cards. Also, with our corporate financial partnerships avail education loans at 0% interest rate

Razorpay PropelId

The IoT Academy
Connecting The Unconnected

Placement Partner

uct

Payment Partner

What's Included in the Fee

- 250+ Hours of Instructor Led Live Training
- Certification from E&ICT Academy, IIT Roorkee
- Learn From IIT Professors and the Industry Experts
- Access to Course Materials and Resources
- 50+ Hands-on Projects and Case Studies
- 100% Support by the Dedicated Industry Partner Team

Additional Benefits

- Lifetime Access to Live Recorded Sessions
- Complimentary Career Guidance or Resume-building
- 3 Months Internship Experience By Industry Partners
- 1:1 Doubt Resolution In Live Sessions

Tools

jupyter | tableau | Midjourney | ChatGPT | DALL·E | Gemini | Google Cloud AutoML Vision | Amazon SageMaker | GitHub | kaggle | Power BI | perplexity | deepseek | git | OpenAI

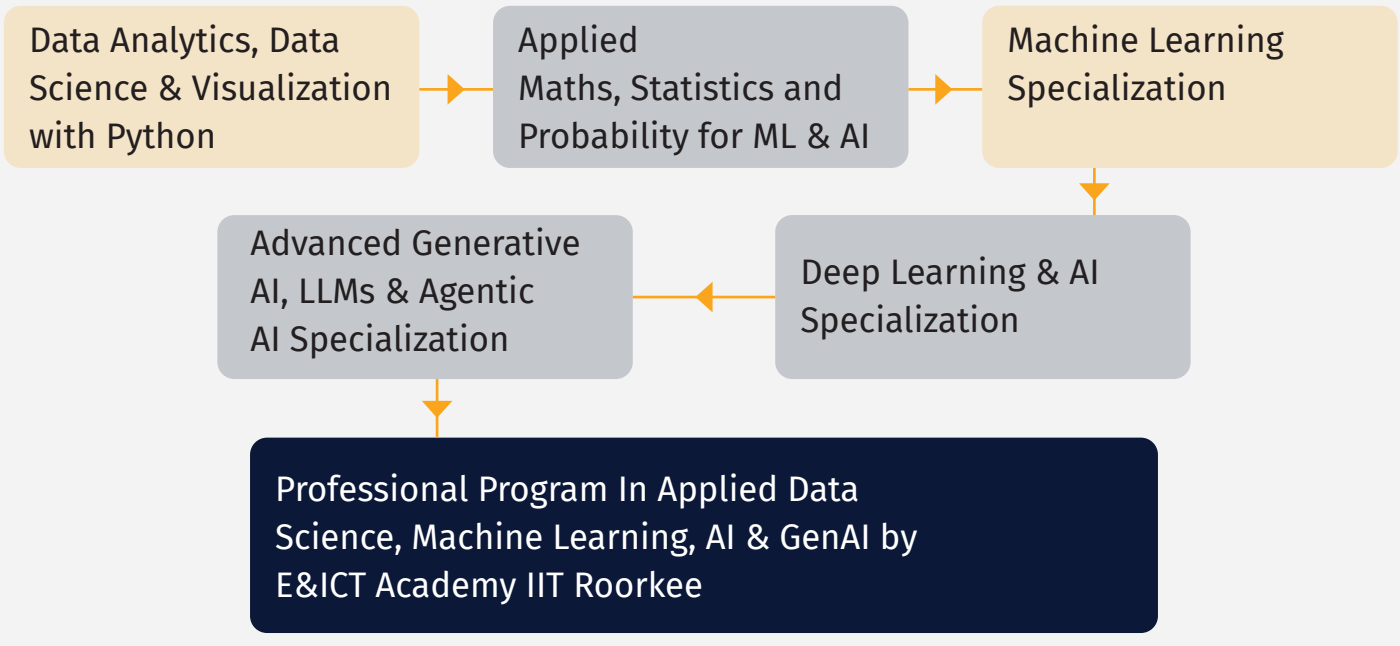
Languages

python | SQL | R

Frameworks

pandas | bokeh | matplotlib | OpenCV | spaCy | NumPy | scikit-learn | PyTorch | TensorFlow | MySQL | Keras | Hugging Face | TensorFlow Lite

Learning Path



Track Curriculum

Module 1 Module 2 Module 3 Module 4 Module 5

Data Analytics, Data Science & Visualization with Python (48 hours 16 Session)

Python Basics

- What is Python
- Application of Python
- Why use Python for AIML Installation
- Anaconda/Other Idle
- Idle to IDE
- Lambda Functions
- Data Types in Python
- Conditional Statement
- Loops in Python
- Functions in Python
- Advance Functions
- File Handling
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

Intro to Git & Github and Kaggle

- Git and GitHub
- kaggle

Module 1 Module 2 Module 3 Module 4 Module 5

Python NumPy & Pandas

- Introduction to NumPy
- Exploring a NumPy Array
- Indexing & Slicing a NumPy Array
- Manipulating a NumPy Array
- Performing Mathematical & Statistical Functions using NumPy
- Performing Linear Algebra Operations using NumPy
- Introduction to Pandas
- Exploring Pandas Series
- Introduction to Pandas DataFrame
- Importing & Exporting Data
- Implementing basic DataFrame functionalities
- Exploring Descriptive Statistics with Pandas
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

R Language Essentials

- Fundamentals of R
- Vectors & control statements
- Functions in R
- Matrices & strings
- Lists & arrays in R
- Data visualization in R

Module 1 Module 2 Module 3 Module 4 Module 5

Exploratory Data Analysis (EDA)

- Data Types
- Dispersion & Skewness
- Data imputation
- Data Pre-processing
- Data Cleaning
- Data Manipulation
- Advanced Manipulation
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

SQL For Data Analytics

- SQL Basic
- SQL Joins
- SQL Aggregations
- Subqueries and Temp Tables
- SQL Data Cleaning
- Window Functions
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

Data Analysis with Excel

- Key Formulas and Functions, Ranges and Tables
- Data Cleaning – Text Functions, Dates and Times
- Conditional Formatting, Sorting and Filtering
- Dashboard Creation
- Analysis with Pivot Tables
- Data Analysis in Excel – Trends and Patterns
- Data Visualization in Excel – Charts and Plots
- Functions in Python
- Advance Functions File Handling
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

Data Visualization with Python

- Why Data Visualization?
- Introduction to Data Visualization
- Libraries & Tools for Data Visualization in Python
- Static Data Visualization Using Seaborn
- Interactive Data Visualization Using Plotly Express
- Interactive Animations & Facet Plots

Module 1 Module 2 Module 3 Module 4 Module 5

Tableau for Business Intelligence

- Understanding Data
- Creating Your First visualization
- Tableau Calculations
- Formatting Visualizations
- Manipulating Data in Tableau
- Creating Dashboards AND Stories
- Distributing & Publishing Your Visualization
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

Visual Storytelling using Power BI

- Introduction To Power BI
- Creating Power BI Reports, Auto Filters
- Report Visualization And Properties
- Chart And Map Report Properties
- Hierarchies And Drilldown Reports
- Power BI Deployment & Cloud
- Improving Power BI Reports And More...
- **Mini Project**

Capstone Project- 1 in Data Analytics

Module 1 **Module 2** Module 3 Module 4 Module 5

Applied Maths, Statistics and Probability for ML & AI (21 Hours 7 Sessions)

Mathematics for Machine Learning & AI

- Linear Algebra
- Multi-variable Calculus
- Introduction to Calculus
- Scalar and Vectors
- Text Embeddings
- Gradient Descent

Module 1 **Module 2** Module 3 Module 4 Module 5

Statistics for Machine Learning & AI

- Applications of Statistics
- Introduction to Statistics
- Categories of Data
- Basic Terminologies in Statistics
- Sampling techniques
- Descriptive Statistics
- Measure Used in Descriptive Statistics
- Z-Scores
- Measures of central Tendency
- Feature Scaling
- Skewness

Module 1 **Module 2** Module 3 Module 4 Module 5

Probability

- What is Probability?
- Rules of probability
- Types of Probability
- Random Variables
- Probability Distribution Functions
- Mini Project

Module 1 **Module 2** Module 3 Module 4 Module 5

Inferential Statistics

- Introduction to Inferential Statistics
- Hypothesis Testing
- Normal Distribution
- P-value
- One-tailed and Two-tailed tests
- One Sample Z test
- One Sample T test
- Independent Sample T test
- Chi-square test
- ANOVA

Capstone Project- 2

Module 1 Module 2 **Module 3** Module 4 Module 5

Machine Learning Specialization (60 Hours 20 Sessions)

Introduction to Machine Learning

- What is Machine Learning?
- Applications of Machine Learning
- Machine Learning in your daily life
- Machine Learning in Retail
- Steps Involved in Machine Learning

Module 1 Module 2 **Module 3** Module 4 Module 5

Regression

- Introduction to Regression
- Linear Regression
- Evaluation Metrics in Regression Models
- Logistic Regression
- **Mini Project**

Supervised Classification

- Why Use Classification?
- Application of Classification Algorithms
- Introduction to Classification
- Types of Classification Algorithms
- Classification: Decision Tree
- Classification: Random Forest
- ML in Banking & Finance - Benefits
- Classification: SVM
- Classification: KNN
- Classification: Naïve Bayes
- Evaluating Classification Models
- Model Optimization Techniques
- Model Boosting Techniques
- Introduction to PyCaret
- Dealing with Unbalanced Datasets
- **4 Mini Projects**

Capstone Project- 3

Module 1 Module 2 **Module 3** Module 4 Module 5

Unsupervised Learning

- What is Unsupervised Learning?
- Application of Unsupervised Learning
- Introduction to Clustering
- Types of Clustering
- Partitioning Methods: K-means, DBSCAN, Spectral
- Hierarchical Methods: Hierarchical
- **2 Mini Projects**

Module 1 Module 2 **Module 3** Module 4 Module 5

Dimension Reduction

- PCA
- Factor Analysis
- LDA
- **Mini Project**

Module 1 Module 2 **Module 3** Module 4 Module 5

Association Rules Mining

- What are Association Rules?
- Association Rule Parameters
- A-priori Algorithm
- Market Basket Analysis

Module 1 Module 2 **Module 3** Module 4 Module 5

Recommendation System

- What is a Recommendation System?
- Need for a Recommendation System
- Recommendation System Use Cases
- Applications of Recommendation System
- Types of Recommendation Systems
- Collaborative Filtering
- Content Based Filtering
- Matrix Factorization
- Pros and Cons of Collaborative Filtering
- Hybrid Recommender System
- **Mini Project**

Module 1 Module 2 **Module 3** Module 4 Module 5

Time-series Forecasting

- Introduction to forecasting data
- Properties of Time Series data
- Features of Time Series data
- Markov Processes - Overview and Terminologies
- Naive, Average and Moving Average Forecasting
- Exponential Smoothing
- ARIMA Approach
- **Mini Project**

Module 1 Module 2 **Module 3** Module 4 Module 5

Machine Learning Model Deployment

- Overview of Machine Learning Models
- Machine Learning System Architecture
- Research Environment
- Packaging, Serving and Deploying the model
- Differential Testing And More...

Capstone Project- 4

Module 1 Module 2 Module 3 **Module 4** Module 5

Deep Learning & AI Specialization (57 Hours 19 Sessions)

Introduction to Deep Learning

- What is Deep Learning
- Curse of Dimensionality
- Machine Learning vs. Deep Learning
- Use Cases of Deep Learning
- Human Brain vs. Neural Network
- What is Perceptron?
- Learning Rate
- Epoch
- Batch Size

Module 1 Module 2 Module 3 **Module 4** Module 5

Computer Vision

- Introduction to Convolutional Neural Networks
- Introduction to images
- Convolution, Pooling, Padding & its mechanisms
- Forward Propagation & Backpropagation for CNNs
- CNN architectures
- AlexNet, VGGNet, InceptionNet & ResNet
- Transfer Learning
- Object Detection

Module 1 Module 2 Module 3 **Module 4** Module 5

Introduction to NLP

- Introduction to NLP
- Libraries & Tools for NLP in Python
- NLTK vs Spacy
- Applications of NLP:
- Chatbot
- Search, Autocorrect and Autocomplete
- Grammar Checker

Module 1 Module 2 Module 3 **Module 4** Module 5

Text Processing Methods

- Bag of Words
- Countvectorizer
- Term Frequency (TF)
- Inverse Document Frequency (IDF)
- Converting text to features and labels
- Multinomial Naive Bayes Classifier
- Leveraging Confusion Matrix Assignment
- Word Embeddings
- Word2Vec
- **Mini Project**

Module 1 Module 2 Module 3 **Module 4** Module 5

TensorFlow Hub for Object Detection using Faster RCNN

- Introduction to TensorFlow Hub
- Use cases of TensorFlow Hub
- Limitations of CNN in object detection
- Architecture of RCNN
- Applications of RCNN
- Types of RCNN
- Step by step implementation of Faster RCNN

Module 1 Module 2 Module 3 **Module 4** Module 5

Sentiment Analysis

- Sentiment Analysis
- Subjectivity Analysis
- Topic Extraction
- Product Reviews
- Opinion Retrieval and Spam
- Opinion Summarization
- Implementing Sentiment Analysis in Python

Module 1 Module 2 Module 3 **Module 4** Module 5

Reinforcement Learning

- Understanding Reinforcement Learning
- Algorithms associated with RL
- Q-Learning Model
- **Mini Project**

Module 1 Module 2 Module 3 **Module 4** Module 5

MLOps/LLMOps & Deployment

- Introduction To MLOps
- Automating Pipelines with Airflow
- Model Packaging
- API Building
- Model Training with AWS SageMaker
- Model Deployment
- Model Maintenance
- Git, Github and Github Actions
- CICD Pipeline
- Model packaging via Dockers and containers
- Deployment on DataBricks

Capstone Project- 6

Module 1 Module 2 Module 3 Module 4 **Module 5**

Advanced Generative AI, LLMs & Agentic AI Specialization (54 Hours 18 Sessions)

Introduction To Generative AI

- Introduction to Generative AI Models
- The Future of Generative AI
- Types of Generative AI Models
- Ethical Considerations in Generative AI Models & ChatGPT
- Popular Generative AI Models
- Benchmarking & Evaluating Models

Module 1 Module 2 Module 3 Module 4 **Module 5**

ChatGPT, Prompt Engineering, Explainable AI ChatGPT

Fundamentals of Prompt Engineering

- Zero-shot, Few-shot, Chain-of-Thought (CoT) Promptin
- Role-based Prompts, Multi-turn Conversations

Advanced Prompt Techniques

- System vs. User Prompts, Style Transfer, Iterative Prompting

Explainable AI

- Introduction to explainable AI
- Mini Project

Module 1 Module 2 Module 3 Module 4 **Module 5**

Advanced Generative AI Models & Architecture

- Introduction to Generative Models
- Large Language Models (LLMs)
- LLMs: GPT-4, DeepSeek, LLaMA, Mistral, Claude, Falcon
- Variational Autoencoders (VAEs)
- Generative Adversarial Networks (GANs) - StyleGAN, CycleGAN
- Attention Mechanisms and Transformers (GPT, BERT, T5)
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 **Module 5**

Advanced Generative AI - Building LLM Applications

- Langchain and Workflow Design
- Advanced Prompt Engineering Techniques
- LLM Application Development
- LLM Fine-Tuning and Customization
- Fine-Tuning & Pre-training of LLMs
- Benchmarking and Evaluation of LLM Capabilities
- LangChain for LLM Development
- RAG with Hugging Face & LangChain
- **2 Mini Projects**

Module 1 Module 2 Module 3 Module 4 **Module 5**

Agentic AI & Multimodal AI (Voice, Video & Document AI)

Agentic AI

- Frameworks
- Hugging Face Ecosystem
- Speech-to-Text & AI Voice Processing
- Whisper, Wav2Vec, DeepSeek-STT

Multimodal AI Text-to-Image & Image-to-Image Generation

- DALL·E, MidJourney, and Stable Diffusion
- LDMs & Hugging Face Implementations

Advanced Multimodal Models

- Google's Gemini, OpenAI's GPT-4o, and models from Meta AI
- Running ollama locally
- Deployemnt with FastAPI and Model Evaluation
- What MCP is and Why we need it?
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 **Module 5**

Generative AI Applications In Industries

- Applications in Healthcare (e.g., Medical Imaging, Diagnosis)
- Applications in Finance (e.g., Risk Modeling, Fraud Detection)
- Applications in Retail and E-commerce (e.g., Personalized Marketing)
- Applications in Media and Entertainment (e.g., Content Creation)

Module 1 Module 2 Module 3 Module 4 **Module 5**

Responsible AI

- Awareness, Toxicity, Misinformation, Bias, Security, Fairness

Module 1 Module 2 Module 3 Module 4 **Module 5**

No-Code/Low-Code Agentic AI & AI Automation

- Introduction to AI Automation & No-Code AI
- n8n Fundamentals & Workflow Design
- AI-Powered Workflows using n8n
- LLM Integration with n8n
- AI Agents & Tool Calling in n8n

AI tools Covered

GitHub Copilot | CURSOR | Copilot

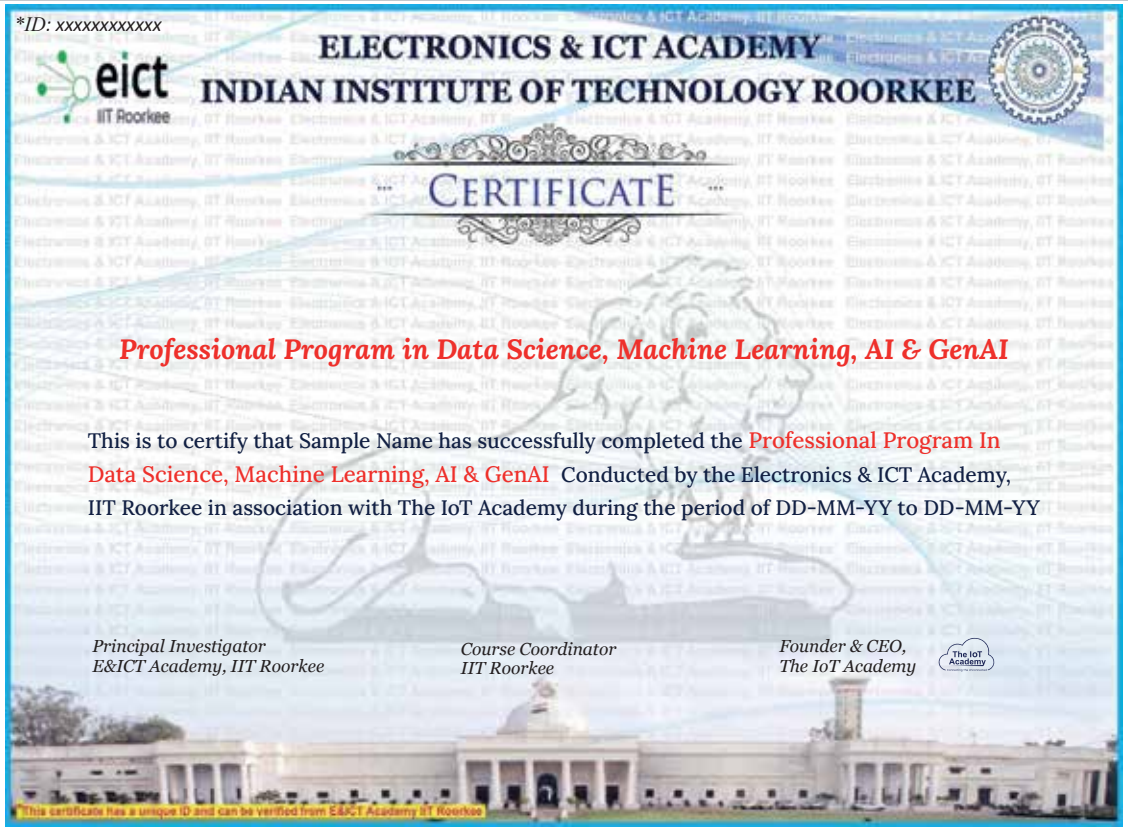
Gamma | Gemini | Lovable

Certification

Upon successfully completing this program, you'll earn certificate in Professional Program In Data Science, Machine Learning, AI & GenAI that is co-branded by E&ICT Academy, IIT Roorkee.

This certificate will testify to your skills as an expert in Data Science, Machine Learning, AI & GenAI.

SAMPLE CERTIFICATE





Industry Based Capstone & Live Session Projects

The projects and assignments will help you accumulate real-world experience in different industries.

- Automotive Industry
- Banking Sector
- Healthcare Industry
- Retail Industry
- Finance Sector
- Educational Industry
- Entertainment Industry
- Manufacturing Segment
- Production industry



Capstone Projects

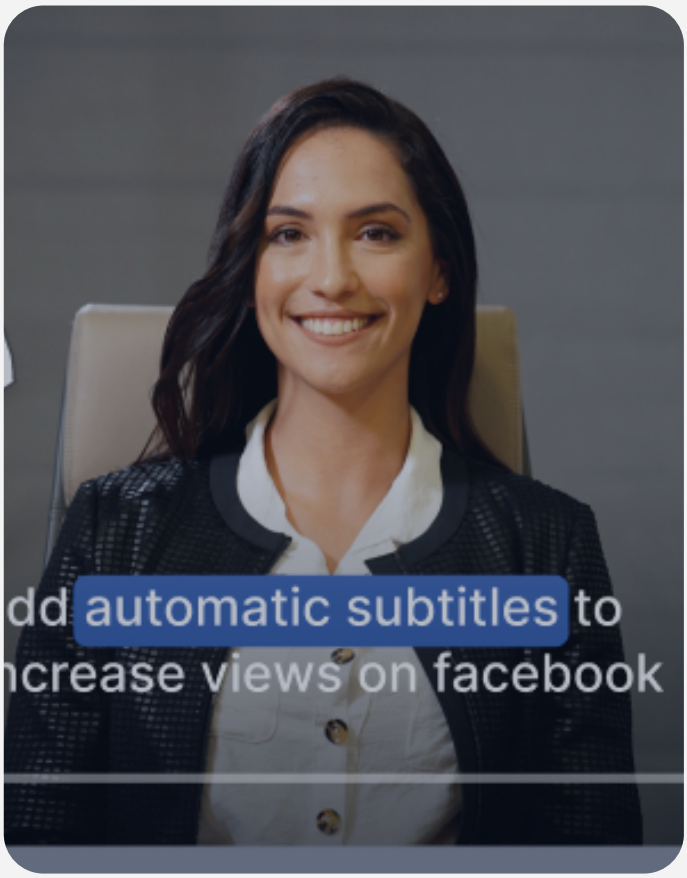
Industry projects will be a part of your Professional Program In Data Science, Machine Learning, AI & GenAI to consolidate your learning. Industry projects will ensure you have real-world experience to start your career in DS, ML, AI & Gen AI.

- **10+ Essential Tools**
- **Designed by Industry Experts**
- **Get Real-world Experience**

Auto Image Captioning

Automatic image captioning is widely used by search engines to retrieve and show relevant search results to the users. For example - to categorize personal multimedia collections, for automatic product tagging in online catalogs, and other areas of business and researches. Use CNN and LSTM to create a model that can automatically add captions to the image.

Tools you will be using: OpenCV, Tensorflow2/ Keras, Numpy, Pandas, Matplotlib





Big Mart - Customer Segmentation

The data scientists at Big Mart have collected 2013 sales data of 1559 products across 10 stores in different cities. Big Mart CEO wants to understand the customer demographics and customer retention (Customer who can converge easily) so that the marketing team can market their products and services by conducting various strategies accordingly.

Tools you will be using: Pandas, scipy, Numpy



Building a RASA Based Chatbot

Rasa is a framework for developing AI-powered, industrial grade, powerful chatbots. The developers use it to create intent-based chatbots. In this project, we are going to understand some of the most importantbasic aspects of the Rasa framework and use RASA NLU and RASA CORE to build a conversational chatbot.

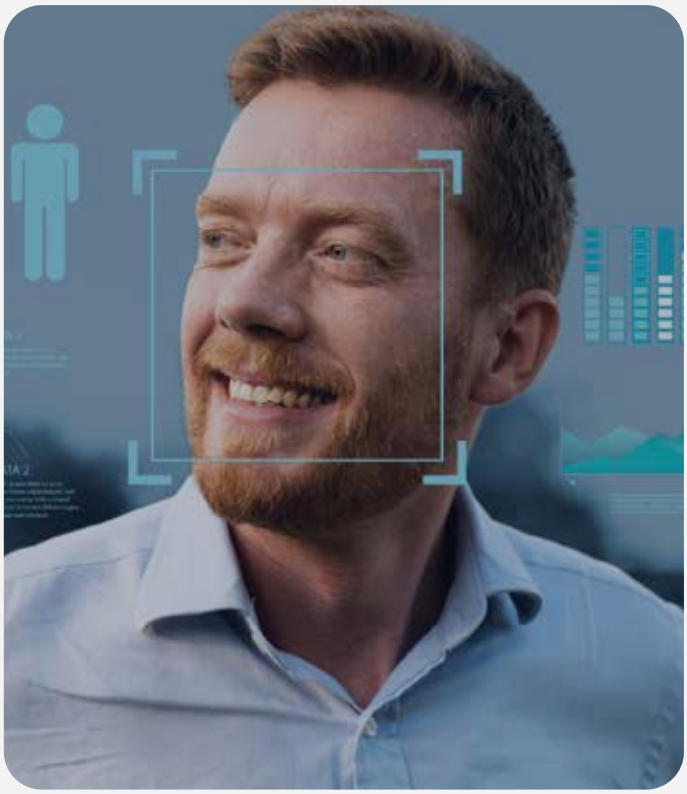
Tools you will be using: RASA NLU, RASA Core



Real-Time Age, Emotion and Gender Detection using CNN

Use CNN and OpenCV to create a model which would detect the person's age, emotion, and age in real-time.

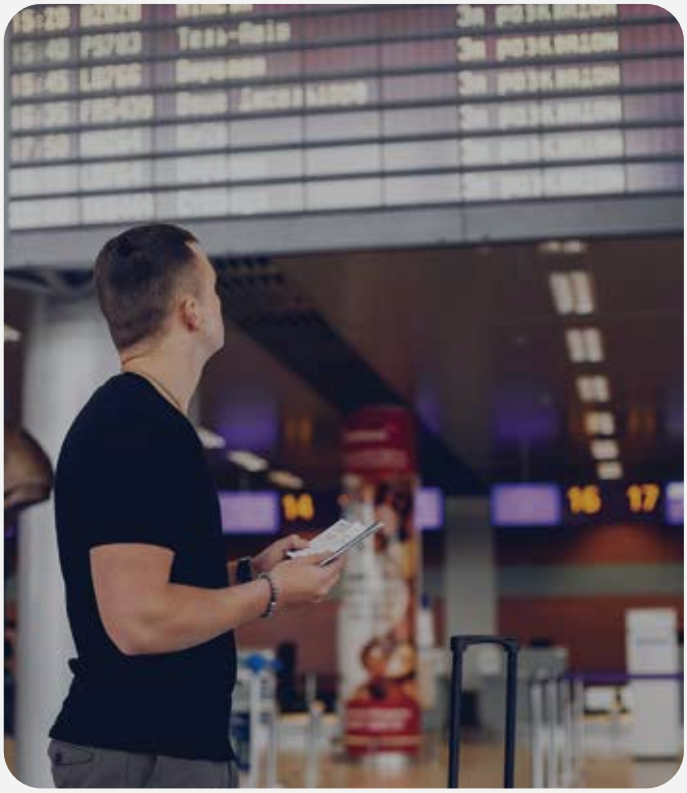
Tools you will be using: OpenCV, Tensorflow2/ Keras, Numpy, Pandas, Matplotlib



Forecast Air Passenger Traffic

An Airline called Star Air has the data of its passengers across months. The data is classified in date/time and the passengers travelling per month. Build a model to forecast the demand (passenger traffic) in Airplanes. You will learn to use Pandas, Scipy, Numpy with hands-on experience of other tools, features and libraries.

Tools you will be using: Pandas, scipy, Numpy



Identify fraudulent credit card transactions

Credit card fraud happens when a fraudster or a thief steals your credit card or the information from that card to make unauthorized purchases in your name or take out cash advances using your account. Credit card companies such as Citibank, HSBC, and American Express need to recognize fraudulent credit card transactionsso that customersare not charged for items that they did not purchase.

Tools you will be using: Sklearn, PyCaret, Streamlit, ELI5, Pandas-Profiling



Traffic Sign Classification using CNN in Tensorflow 2.0

Detection and recognition of traffic signs are crucial for the development of self- driving cars, which have a direct impact on driving behaviors. You will learn to build a CNN model using OpenCV, TensorFlow2, Keras, Numpy, Pandas, Matplotlib to detect and classify the traffic signals for new self-driving cars.

Tools you will be using: OpenCV, Tensorflow2/ Keras, Numpy, Pandas, Matplotlib



Text-to-Image Generation

Description: Combine the power of NLP and CV to generate realistic images from textual descriptions, such as creating visual prototypes for design concepts or generating fictional landscapes.

Tools and Technologies: PyTorch, TensorFlow, OpenCV, Diffusion Models, GANs, Transformers.

Outcomes: Learn to preprocess text and image data, design transformer-based models, and implement diffusion models for image synthesis.





Interactive Chatbot for Healthcare

Description: Build a conversational agent using GPT-based LLMs that assists users with healthcare queries, such as symptom checks or medication reminders.

Tools and Technologies: Rasa, PyTorch, Hugging Face Transformers, Streamlit, NLP Libraries (spaCy, NLTK).

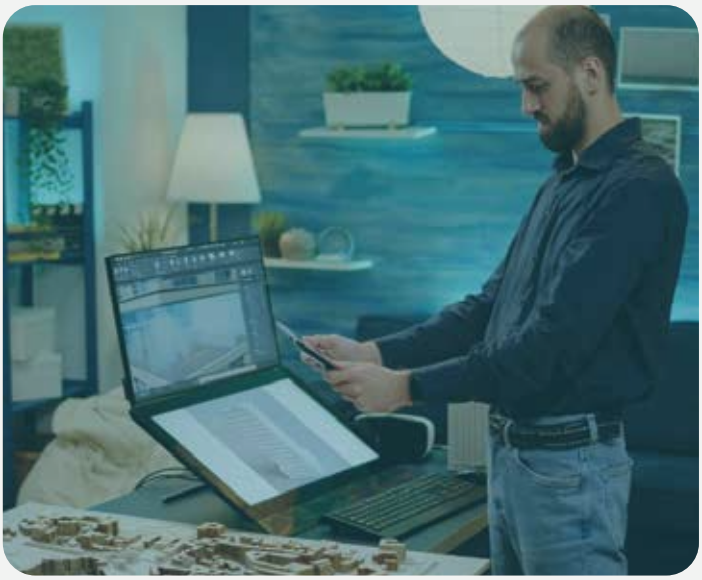
Outcomes: Gain experience fine-tuning LLMs for domain-specific tasks, handling conversational context, and deploying chatbots.

Generative Art Creation

Description: Develop a system to generate artwork using GANs or style transfer techniques. The model will create unique artistic styles or merge different styles into new creations.

Tools and Technologies: TensorFlow, PyTorch, StyleGAN, OpenCV.

Outcomes: Understand GAN architecture, apply advanced loss functions, and explore applications in digital art and design.





Predictive Maintenance with Generative Models

Description: Use Variational Autoencoders (VAEs) to detect anomalies in industrial sensor data and predict maintenance needs.

Tools and Technologies: PyTorch, TensorFlow, VAE, Pandas, NumPy.

Outcomes: Learn latent space representation, anomaly detection, and how generative AI can enhance predictive maintenance.

AI-Powered Content Generation for Marketing

Description: Create a generative AI model to assist marketing teams in generating product descriptions, blog posts, and ad copies.

Tools and Technologies: GPT, Hugging Face, PyTorch, TensorFlow.

Outcomes: Work with fine-tuning strategies for LLMs, data cleaning, and evaluating text quality using BLEU and ROUGE metrics.





Voice Cloning and Audio Synthesis

Description: Develop a model to synthesize human-like speech or clone specific voices using generative models like WaveNet or Tacotron.

Tools and Technologies: TensorFlow, PyTorch, Librosa, WaveNet.

Outcomes: Learn to preprocess audio data, generate waveforms, and build real-time voice applications.

Customer Churn Early-Warning Engine

This project focuses on predicting customer churn by analyzing behavioral and transactional data to enable early intervention and retention strategies.

Tools you will be using:
Python, Pandas, Scikit-learn, SQL, Machine Learning Models



Viral Content Intelligence System

This project analyzes social media content and audience engagement data to identify patterns that contribute to high-performing and viral content.

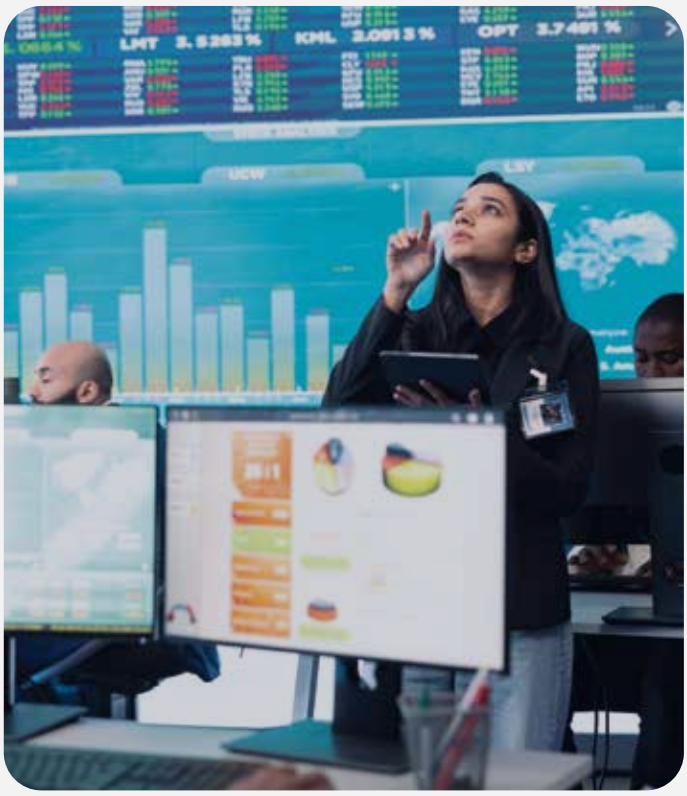
Tools you will be using:
Python, Social Media APIs, NLP Libraries



Quant Risk & Market Volatility Intelligence System

This project analyzes market data to measure financial risk, volatility, and price fluctuations, supporting quantitative and risk-aware investment decisions.

Tools you will be using:
Python, Pandas, NumPy, Time Series Analysis, Financial Analytics Libraries



Fair Pay, Bias & Workforce Ethics Analyzer

This project analyzes compensation and workforce data to detect pay gaps, bias patterns, and ethical risks, promoting fairness and transparency in organizations.

Tools you will be using:
Python, Pandas, Statistical Analysis, Data Visualization Tools



Behavioral & Personality Intelligence Model

This project analyzes user behavior data to model personality traits and behavioral patterns for enhanced personalization and decision-making insights.

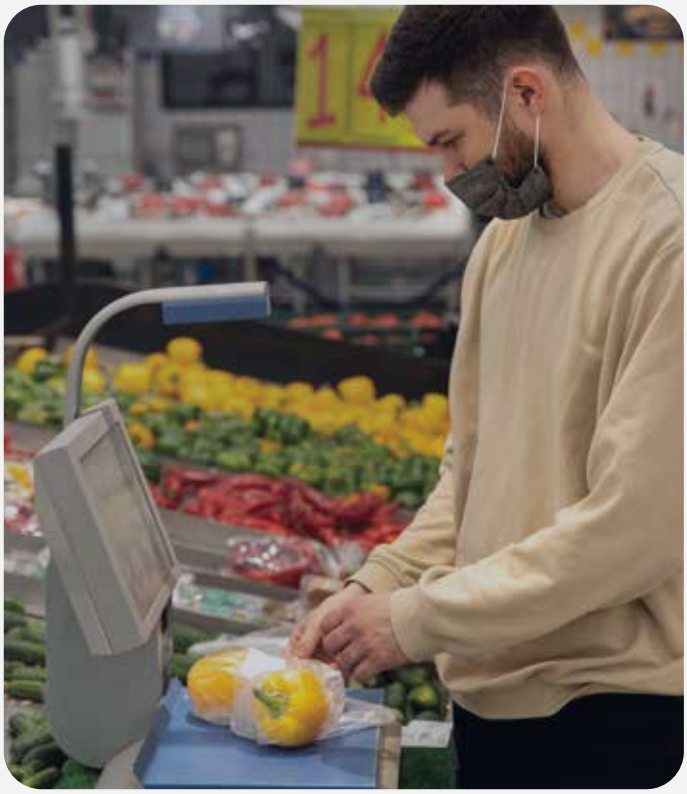
Tools you will be using:
Python, Pandas, NLP Libraries, Machine Learning Techniques



Retail Cross-Sell Recommendation Engine

This project builds a recommendation system to identify cross-sell opportunities by analyzing customer purchase history and product relationships.

Tools you will be using:
Python, Pandas, Recommendation Algorithms, Collaborative Filtering



Socioeconomic Segmentation & Inequality Mapping System

This project analyzes socioeconomic data to segment populations and visualize inequality patterns across regions for data-driven policy insights.

Tools you will be using:
Python, Pandas, GIS Tools, Data Visualization Libraries



Multimodal Sentiment & Emotion Intelligence System

This project analyzes text, images, and videos from social media to detect sentiments and emotions, helping understand public opinion and user behavior.

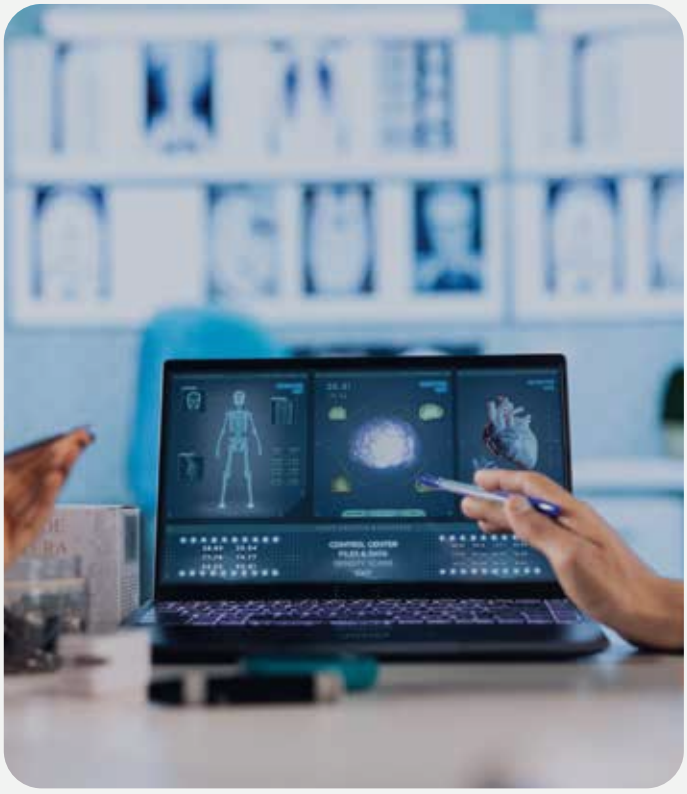
Tools you will be using:
Python, NLP Libraries, Computer Vision Libraries, Deep Learning Frameworks



AI-Assisted Medical Imaging Diagnosis System

This project uses medical imaging data to assist in disease detection and diagnosis by identifying patterns and anomalies in medical images.

Tools you will be using:
Python, OpenCV, Deep Learning Frameworks, Medical Imaging Libraries



Legal Outcome Prediction & Decision Support AI (Law)

This project analyzes legal case data to predict possible outcomes and provide decision-support insights for legal professionals.

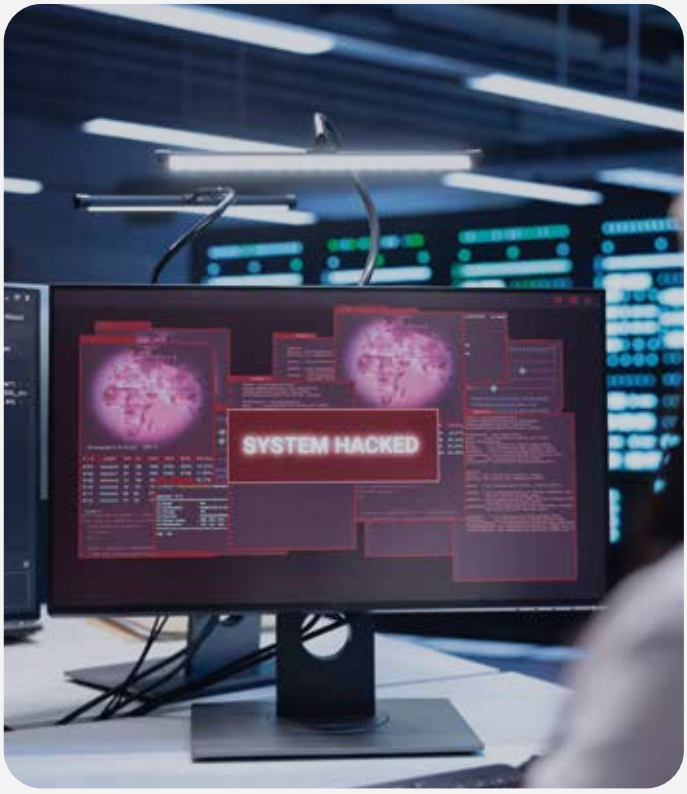
Tools you will be using:
Python, NLP Libraries, Machine Learning Models, Legal Data Sources



AI-Powered Threat Detection & Surveillance Intelligence

This project focuses on detecting potential threats by analyzing surveillance data, enabling proactive security monitoring and risk assessment.

Tools you will be using:
Python, Computer Vision, Deep Learning Models, Surveillance Analytics Tools



Industrial Failure Prediction & Reliability AI System

This project predicts equipment failures by analyzing sensor and operational data to improve system reliability and reduce downtime.

Tools you will be using:
Python, Pandas, Machine Learning Models, Time Series Analysis



Autonomous Marketing Campaign AI Agent

This project builds an autonomous AI agent that plans, executes, and optimizes marketing campaigns by analyzing audience behavior and campaign performance.

Tools you will be using:
Python, GenAI Frameworks, Marketing APIs, Data Analytics Tools



Explainable GenAI Financial Copilot (RAG-Based)

This project develops a GenAI-powered financial assistant using Retrieval-Augmented Generation to provide explainable insights from financial data and reports.

Tools you will be using:
Python, LLMs, RAG Frameworks, Vector Databases, Financial Datasets



Agentic AI Hiring & Interview Automation System

This project automates hiring workflows by screening resumes, scheduling interviews, and generating interview insights using agent-based AI systems.

Tools you will be using:
Python, NLP Libraries, Agentic AI Frameworks, HR Analytics Tools



AI-Driven Drug Discovery Intelligence Agent

This project applies AI techniques to analyze molecular and biomedical data to accelerate drug discovery and candidate identification.

Tools you will be using:
Python, Deep Learning Frameworks, Bioinformatics Tools, Molecular Data Libraries



Multimodal Clinical Documentation & Summarization AI

This project uses multimodal data such as text and voice to generate accurate clinical documentation and automated medical summaries.

Tools you will be using:
Python, NLP Libraries, Speech-to-Text Models, Healthcare Data Tools



GenAI Legal Research & Drafting Copilot

This project builds a GenAI-powered copilot to assist in legal research, case analysis, and automated drafting of legal documents.

Tools you will be using:
Python, LLMs, NLP Libraries, Legal Knowledge Bases



Program Faculty Industry Experts



Kaushlendra Sisodia

Director At Uniconverge Technologies Pvt. Ltd.
IIT Kanpur Alumnus



Mr. Sanjay Aggarwal

Data Science With ML Expert IMS BHU
The IoT Academy



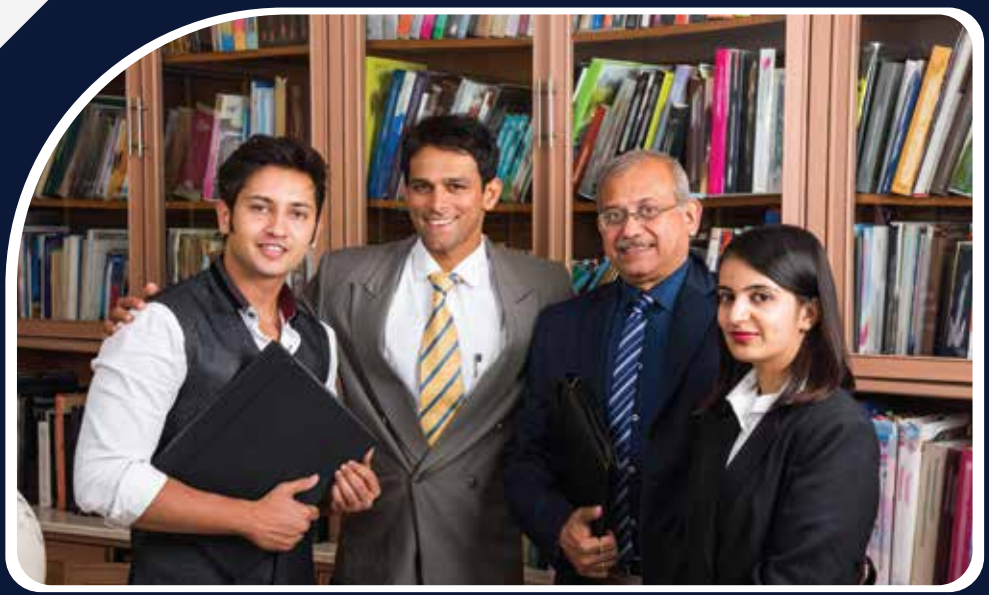
Prof. Sumit Kalra

Department Of Computer Science & Engineering,
IIT Jodpur



Ashwani Kumar

AI Expert 20+ Years Of Experience Of
University Of Miami



Program Eligibility Criteria

Those wishing to enrol in the Professional Program In Data Science, Machine Learning, AI & GenAI By E&ICT Academy IIT Roorkee are required to apply for admission.

For admission in the Professional Program In Data Science, Machine Learning, AI & GenAI, candidates should have:

- Any Undergraduate Degree holder like BCA, B.Tech, B.E, B.Com etc with an average of 50% or higher marks.
- Basic understanding of programming concepts and mathematics. For candidates who do not know Python, we offer a free pre-program tutorial.
- Working Professionals with 2+ years of experience are preferred to apply for this program.

Application Process

The application process consists of four simple steps. An offer of admission will be made to the selected candidates and accepted by the candidates upon payment of the admission fee.

Online Application Form

Apply by filling a simple online application form. You will be required to provide personal, educational, and professional details. Once we receive your details, our Admission Head will reach out to take your candidature further.

Interview Process

Go through a screening call with the Admission Director's office who will gauge your passion and eligibility for the program.

Scholarship & Offer Letter (Provided By Industry Partner)

Apply for the scholarship (not mandatory) and take the test. An offer letter will be rolled out to the selected candidates.

Admission & Batch Allotment

Complete the admission and make a quick block payment formality with assistance from our loan partners, you will be given course credentials and your learning journey will begin!

100% Dedicated Job Support By Industry Partner (The IoT Academy)

Live Career-Oriented Webinars

Live webinar sessions that include curriculum and career services walk throughto help learners understand their learning objective and expectations of hiring managers.

Leadership Skill Development Sessions

Recurring training sessions with experts to help learners Develop Interpersonal & Leadership Skills.

1-on-1 Career Mentoring Sessions

One-on-one Career Mentoring sessions on how to develop the right skills and attitude to secure a dream job.

Exhaustive Interview Preparation

Expert tips, sample interview questions, mock interviews with constructive feedback from industry experts to gain handson experience of technical rounds, HR round, and more.

Job Search Assistance & Job Feeds

Access to multiple job portals to help learners navigate through thousands of jobs including global remote jobs.

Profile Building Assistance

A dedicated Career Coach will provide expert tips on how to create an attractive, relevant resume and LinkedIn profile.



Top Recruiters

Corporates Who Hired form Us



What's In It For You

Top Career Roles You Can Explore After Completing the Program:

AI Research Engineer | Data Analyst | Associate Machine Learning Engineer | Research Analyst

Entry-Level Role	Mid-Level Role	Senior-Level Role
• AI Research Engineer • Data Analyst • Associate Machine Learning Engineer • Research Analyst	• Senior Data Scientist • Machine Learning Research Scientist • Deep Learning Expert • AI/ML Product Lead • AI Solutions Architect	• Chief AI & Data Officer • Head of AI Strategy • Principal Machine Learning Scientist • Director – Generative AI & Innovation

Program Influencer

The best of academics and industry experts to give you a robust learning experience. We aim to give you a complete exposure to real-time and relevant industry applications of the concepts learn.



Mr. Anil Pandey

20+ Yrs Exp, Director of R&D Protocol Stack Development at Rohde & Schwarz MTC, UK, IIT-Kanpur Alumnus



Dr. Ashwani Singh

18+ Yrs Exp, Schneider, Ph.D. France Engineering, Innovation Management, IIoT, Industry 4.0.



Mr. Rajiv Bajpai

20+ Yrs Exp, Apple (USA), ex-Broadcom, ex-Samsung, IIT-Kanpur Alumnus.



Mr. Tej Pratap Pandey

20+ Yrs Exp, COO at Mobiotics, IIT Kanpur Alumnus



Mr. Manbhawan Prasad

20+ Yrs Exp, Product Manager at Microsoft, Ireland EMCC Certified Coach, IIT-Guwahati.



Mr. Ashish Shrivastava

16+ Yrs Exp, Senior System Designer, Algorithm and System, LTE/5G Baseband at Ericsson, Sweden



Mr. Deepak Singhal

20+ Yrs Exp, Enterprise Architect Director at Capgemini, Speakeron Cloud technologies

Notable Alumni Of The IoT Academy

The best of academics and industry experts to give you a robust learning experience. We aim to give you a complete exposure to real-time and relevant industry applications of the concepts learnt.



Rohit Singh
Data Scientist
NeoSOFT TECHNOLOGIES



Mr. Swapnil Bhanushali
Associate
DBS



Gaurav Lad
Sr. Associate Growth & Marketing
cityflo



Anushi Mittal
Risk Management Analyst
IKM



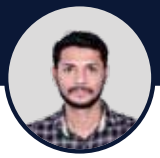
Swati Verma
Jr BI Engineer
ScaleupAlly



Shubham singh
Field Services Engineer



Dr. Hira Zaheer
Assistant Professor of Mathematics



Satyam
I&F Decision Practitioner Senior Analyst
accenture

E&ICT, IIT Roorkee

Professional Program In
Data Science, Machine Learning AI & GenAI

By E&ICT Academy, IIT Roorkee

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